

Driving Assist with Lane Curve Detection

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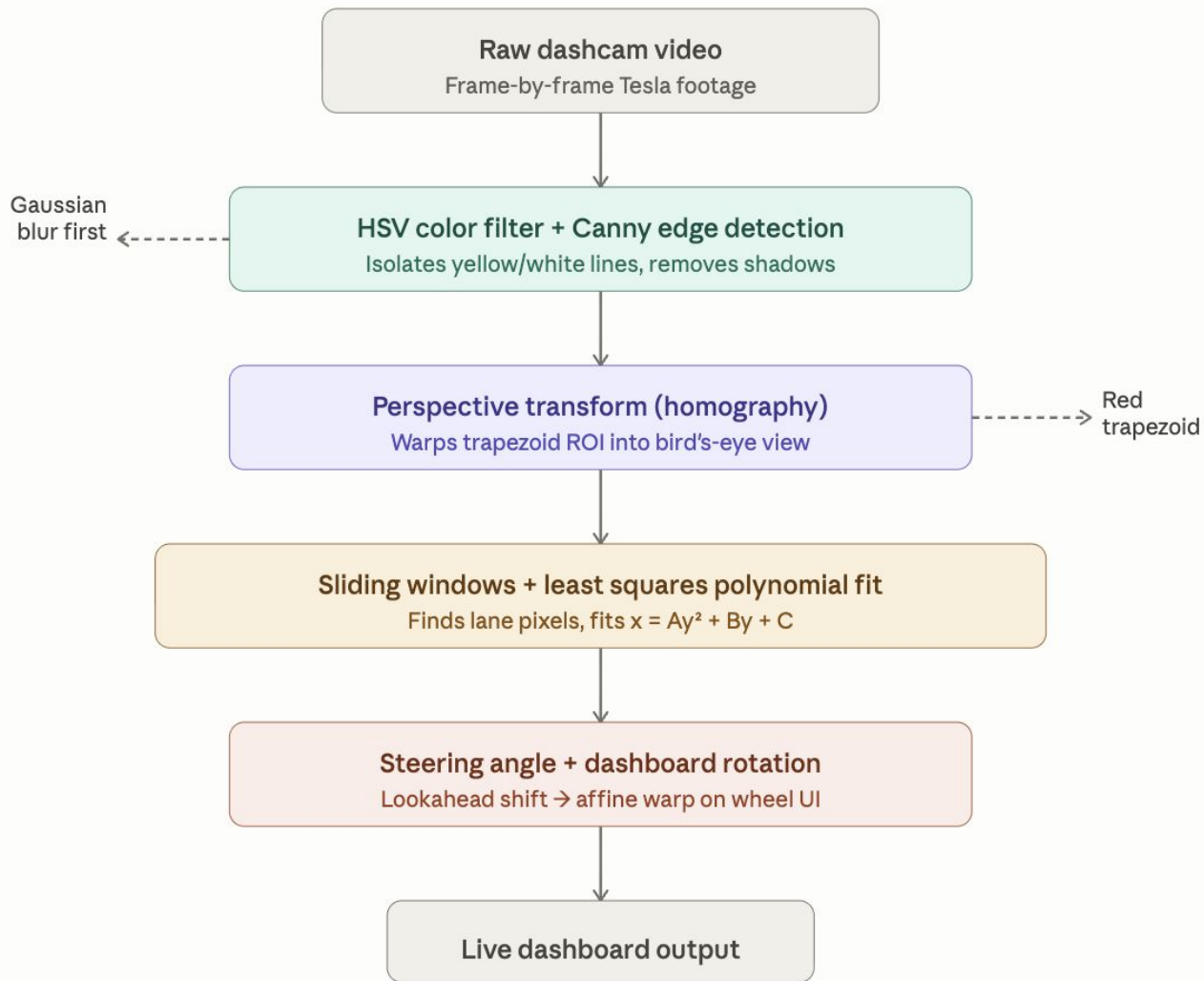
The Goal

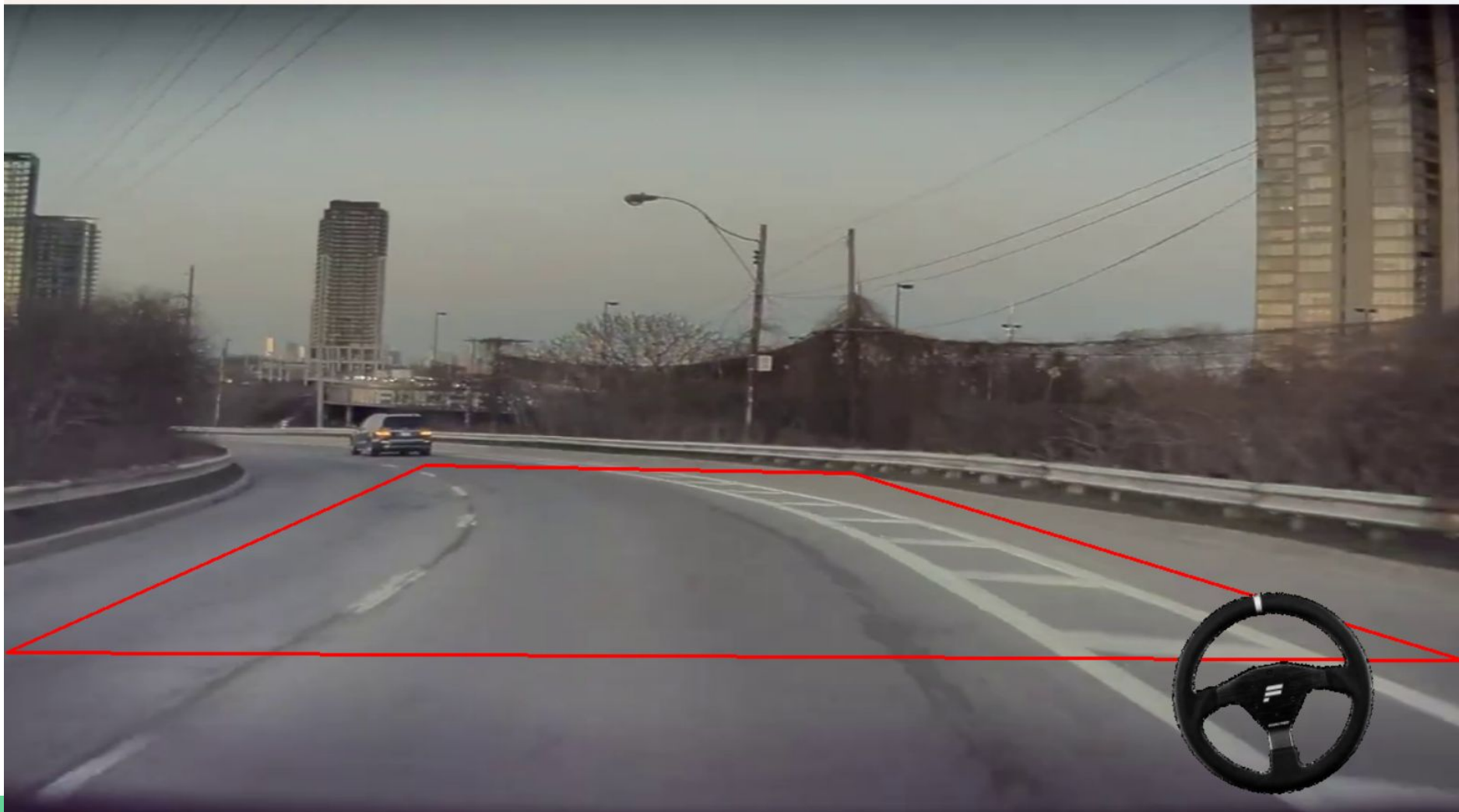
1. Identify road lines
2. Figure out curve of the lane
3. Turn the steering wheel accordingly

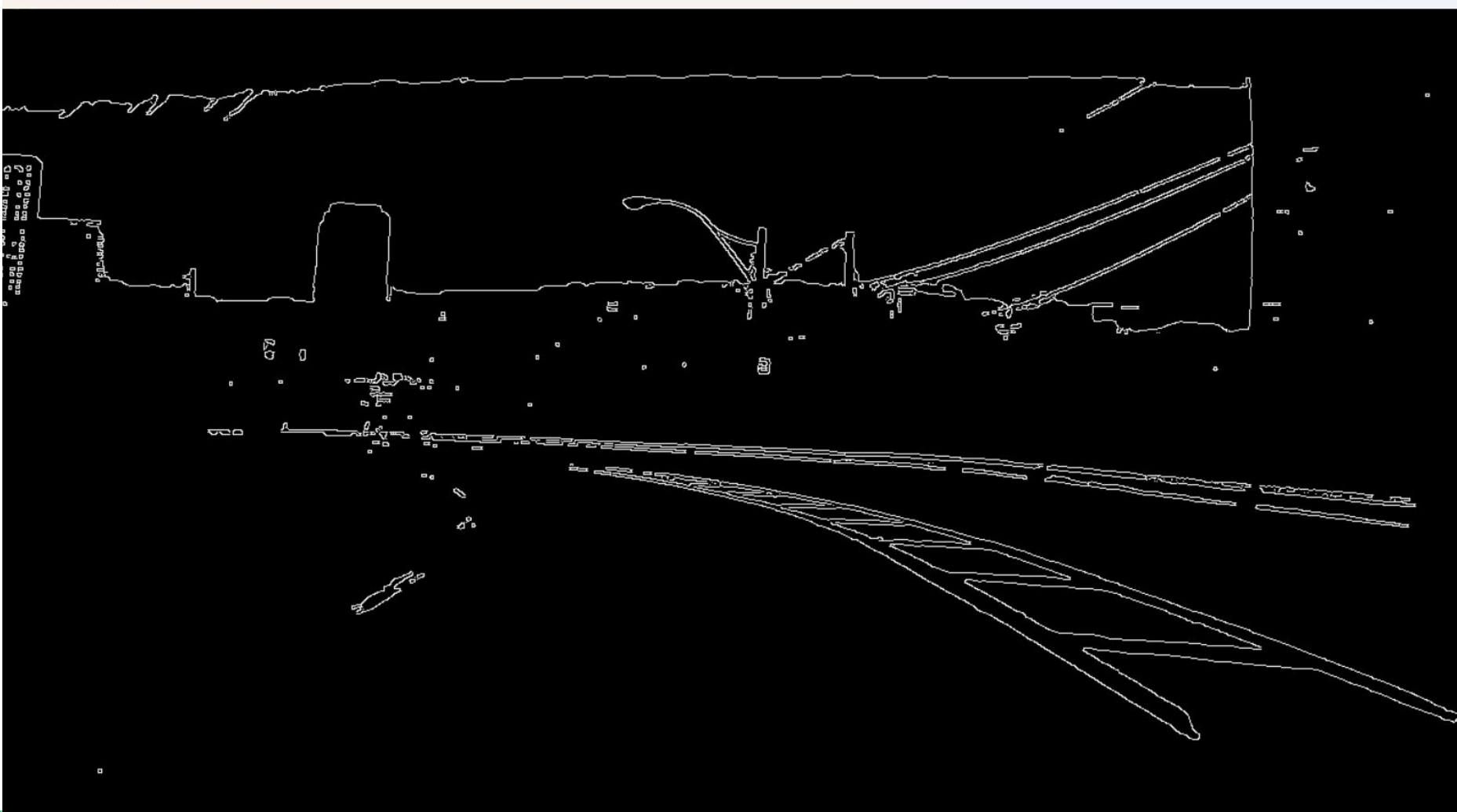


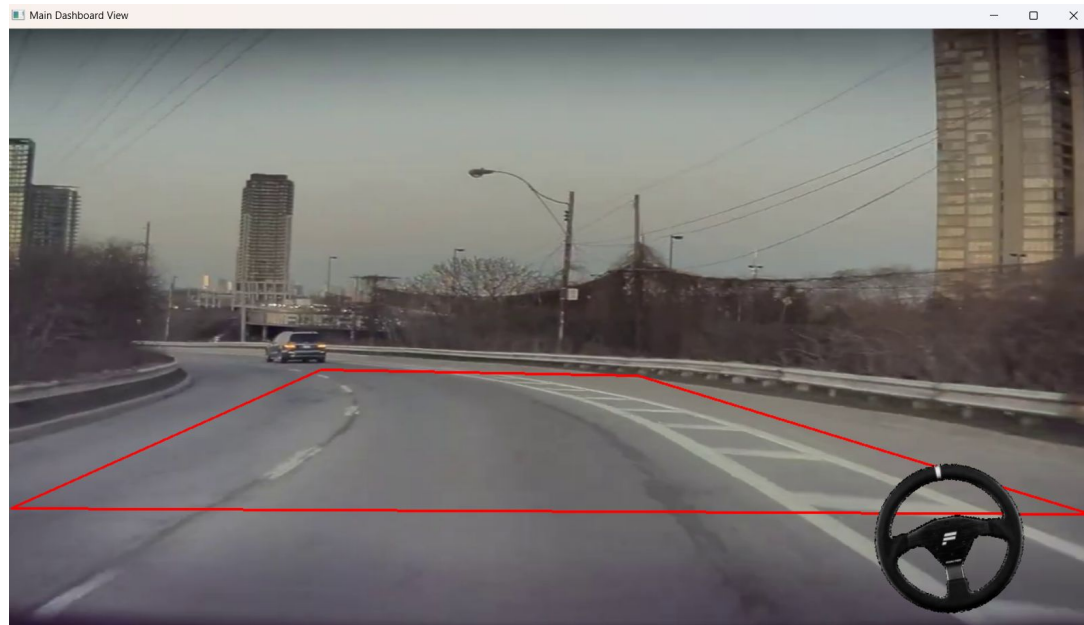
Overview

- Take RAW Tesla video recording
 - Find edges
 - Warp image into a top-down view
 - Find curve of lane
 - Turn steering wheel accordingly
 - Repeat for each frame
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- Input video: Tesla front dashcam footage (raw, unedited, real-world highway driving)
 - Link: <https://www.youtube.com/watch?v=xbPUAb0EeQ4>







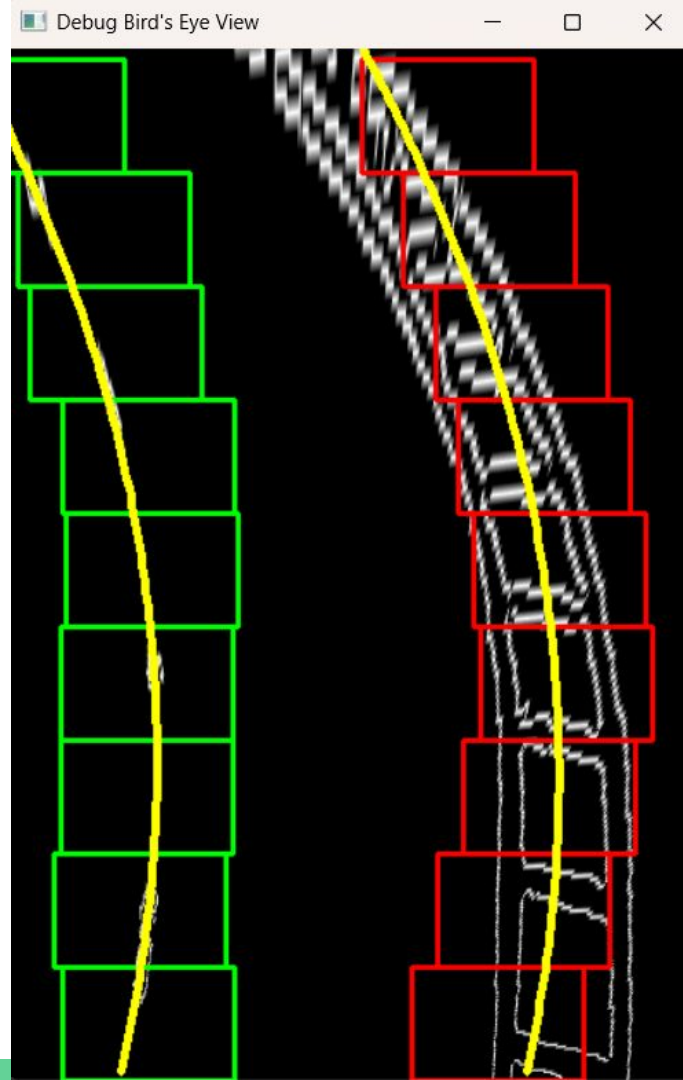


$$x = Ay^2 + By + C$$

A = curvature

B = heading

C = base position



Demo

Improvements to make

- Increase accuracy of curve identification
- Smooth out frame-by-frame wheel turning
- Make trapezoid (ROI) placement automated